

SOFTWARE ENGINEERING LAB

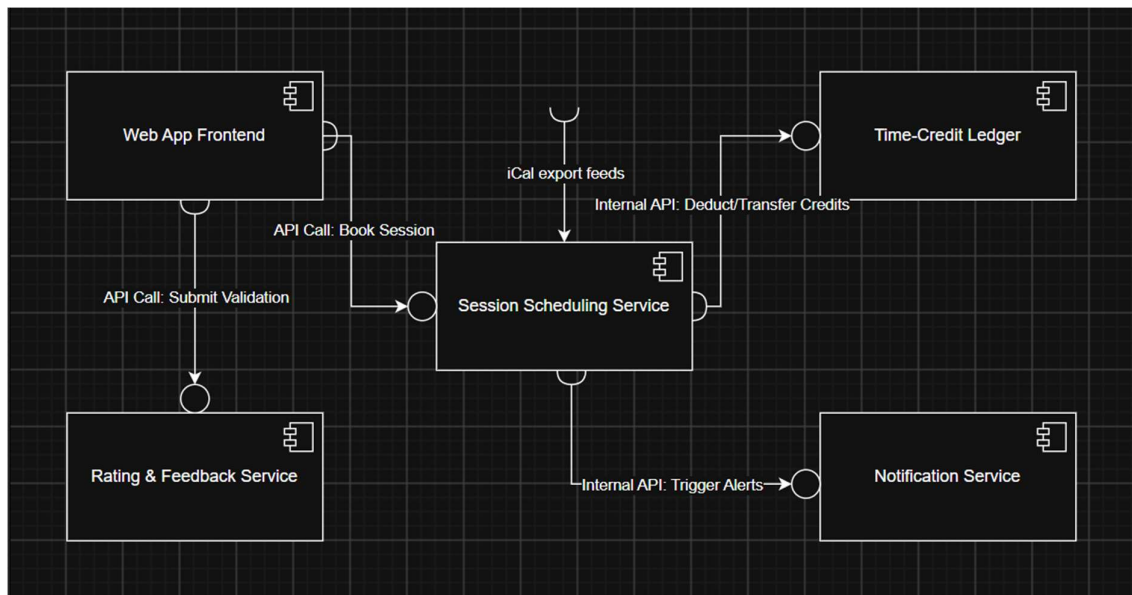
Lab 3: Component Modelling & Architectural Pattern Selection

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1. Component Diagram



Architectural Choice Justification:

- **Domain Isolation:** The platform requires fundamentally different operations, such as managing a highly critical time-credit balance versus handling iCal export feeds for calendar syncing. Microservices allow us to build, deploy, and maintain these functions as separate, fault-isolated components rather than a tightly coupled system.
- **Technology Diversity:** We can deploy specialized database technologies tailored to the specific needs of different services. For example, the Time-Credit Ledger can utilize a highly consistent relational database to guarantee precise transaction

records, while the Rating & Feedback service could utilize a fast, document-based NoSQL database to store text reviews and 1-5 star ratings.

Security Advantage: By isolating the Time-Credit Ledger into its own dedicated microservice, we ensure that potential security vulnerabilities within the public-facing Web App Frontend or the Rating & Feedback Service do not compromise the time-credit balances, which act as the core currency of the platform. Access to the ledger is strictly controlled and obfuscated via internal APIs, keeping critical financial-style data entirely separated from external user interactions.

Performance Benefit: Microservices provide the ability to execute independent scaling for individual components. During periods of high traffic where users are heavily browsing mentor profiles and checking availability, we can horizontally scale the Session Scheduling Service to maintain low latency. This prevents scalability bottlenecks and maintains seamless performance without wasting server compute resources on scaling less frequently accessed components like the Rating & Feedback Service.